

PROJECT PROGRESS REPORT

- Project Name: AirVisionGuard
- Reporting Period: May 2026
- Date of Submission: May 26, 2026
- GitHub: <https://github.com/ThunGg/AirVisionGuard>
- Team Information:

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1. Summary

- Overall Status: On Track
- Abstract: The desktop application is fully functional and operational, with deployment packaging currently pending. Current efforts are now focused on refining and optimizing the AI models.

2. Completed Deliverables

- Pipeline Implementation: Successfully deployed the end-to-end processing pipeline: Camera Input → Image Acquisition → Face Detection → Face Recognition (Embedding generation followed by Cosine Similarity evaluation against anchor references).
- System Infrastructure: Finalized the backend architecture and user interface (UI) development.
- Optimization: Conducted experiments focused on minimizing model power consumption.

3. Evaluation & Key Results

a. Benchmark Environment

All performance and efficiency metrics listed below were benchmarked on an Intel® Core™ i7-8650U CPU @ 1.90GHz.

b. Face Detection & Face Recognition Performance

To evaluate core model efficiency, the following metrics were recorded:

- Disk size: 2.26 MB
- RAM added: 1.34 MB

- FPS: 37.25 frames/sec
- Power: 29.52 W
- Energy/frame: 0.79 J
- Frames/Joule: 1.26 frames/J

c. Authorized Cached Stage

Performance metrics for the integrated system (Application + Face Detection + Face Recognition) under cached conditions:

- Computation Time: 28.47 ms
- System Throughput: 2.00 FPS
- Average CPU Utilization: 2.29%

4. Ongoing Tasks & Next Steps

- Model Efficiency Optimization: Researching and implementing optimization techniques to minimize model power consumption and achieve ultra-low energy overhead.

5. Acknowledgments

- We express our gratitude to SEP and iLab for their support throughout this project phase.